

Assignment 7

Due Friday 10 April at the beginning of class

This Assignment is based on the slides, especially [weeks 10 & 11](#) and [week 12](#).

DO THE FOLLOWING PROBLEMS.

P17. First see the “main idea” about the definition of $C_c^\infty(\Omega)$ in the weeks 10 & 11 slides.

(a) Suppose $\Omega \subset \mathbb{R}^d$ is open. Suppose $g \in L^1_{loc}(\Omega)$ and $\varphi \in C_c^\infty(\Omega)$. Prove that $gD^\alpha\varphi \in L^1(\Omega)$, thus that $\int_\Omega gD^\alpha\varphi \, dm$ is well-defined.

(b) Show that $C(\Omega) \subset L^1_{loc}(\Omega)$. Then, for each dimension d , construct an example of Ω and $f \in C(\Omega)$ showing that “ $C(\Omega) \subset L^1(\Omega)$ ” is false.

P18. Here is an important case using the weak derivative concept in 1D.

Suppose $\Omega = (a, b)$ is a bounded open interval, and let $a = t_0 < t_1 < \cdots < t_n = b$ be a mesh on a closed interval $[a, b]$. As in the slides, the P_1 finite element space is defined as the set of functions which are continuous on $[a, b]$, and which are piecewise linear on each cell (subinterval) $[t_{k-1}, t_k]$. Show that $P_1 \subset H^1(\Omega)$.

P19. *This problem is removed.*

P20. Recall that the norm in $H^k(-\pi, \pi)$ satisfies $(\|f\|_{H^k})^2 = \sum_{\ell=0}^k \int_{-\pi}^{\pi} |f^{(\ell)}(x)|^2 dx$. Here $f^{(\ell)}$ denotes the weak derivative of order ℓ .

(a) Suppose $f \in H^1(-\pi, \pi)$. Using the formulas for Fourier series,¹ prove the derivative formula

$$f'(x) = \frac{1}{\sqrt{2\pi}} \sum_{n \in \mathbb{Z}} in c_n e^{inx},$$

with convergence in L^2 , where c_n are the Fourier coefficients of f .

(b) Show that

$$\|f\|_{H^k}^2 = \sum_{n \in \mathbb{Z}} \left(\sum_{j=0}^k n^{2j} \right) |c_n|^2.$$

(There is no need to justify derivative formulas, beyond what you did in part (a).)

P21. This example asks you to do the details of an example shown in the slides. It shows that H^1 contains discontinuous functions for planar domains. To say it more precisely, there are equivalence classes in H^1 which have no continuous representative.

Let $D = \{x \in \mathbb{R}^2 : |x|^2 < 1/4\}$ be an open disk, centered at the origin, with radius $1/2$. Let $g(x) = \ln(r)$ and $p(x) = \ln(|\ln(r)|)$, in polar coordinates where $|x| = r$, defined

¹See “recall Fourier series in $L^2(-\pi, \pi)$ ” in the [week 12](#) slides.

a.e. on D . (*These functions are not defined at the origin.*) Exactly compute, by hand, the L^2 and H^1 norms of g and p . Then use a computer to plot g and p over $0 < r < 1/2$.

P22. For $z = x + iy$ a complex number, define $p(z) = z^3 - 5z^2 + 2z - 1$. Show that the real and imaginary parts of p , regarded as real-valued functions of (x, y) on $\mathbb{R}^2$, are harmonic.